Optimal control of ODEs
Model Predictive Control

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[**Bibliography**] The following references are related to the lecture:

- Nonlinear Model Predictive Control: Theory and Algorithms, by L. Grüne and J. Pannek, 2011.
- Slides by L. Grüne on "Deterministic Stabilizing and Economic Model Predictive Control", from the 2022 MTNS Conference.

1. INTRODUCTION

[**Continuous-time setting**]
Model Predictive Control (MPC) primarily aims at solving **infinite-horizon** problem of the form:

$$\inf_{(u,y)} \int_0^\infty \ell(u(t), y(t))\,\mathrm{d}t, \quad \begin{cases} \dot{y}(t) = f(u(t), y(t)) \\ y(0) = y_0. \end{cases}$$

These problems are difficult to handle numerically.

Basic idea: solve iteratively **a sequence of optimal control problems on small time horizons**.

Two parameters:

- a sampling time τ
- a prediction horizon T.

[**The method**]
At iteration $k = 0$, do:

- Find a solution $(\bar{u}, \bar{y})$ to

$$\inf_{(u,y)} \int_0^T \ell(u(t), y(t))\,\mathrm{d}t, \quad \begin{cases} \dot{y}(t) = f(u(t), y(t)) \\ y(0) = y_0. \end{cases}$$

- Define $(u_{MPC}(t), y_{MPC}(t)) = (\bar{u}(t), \bar{y}(t))$, for all $t \in (0, \tau)$.

At iteration $k = 1$, do:

- Find a solution $(\bar{u}, \bar{y})$ to

$$\inf_{(u,y)} \int_\tau^{T+\tau} \ell(u(t), y(t))\,\mathrm{d}t, \quad \begin{cases} \dot{y}(t) = f(u(t), y(t)) \\ y(\tau) = y_{MPC}(\tau). \end{cases}$$

- Define $(u_{MPC}(t), y_{MPC}(t)) = (\bar{u}(t), \bar{y}(t))$, for all $t \in (\tau, 2\tau)$.

And so on...

[**The method**]
At the beginning of iteration k, the pair (u_{MPC}, y_{MPC}) is defined on the interval $(0, k\tau)$.
At iteration k, do:

- Find a solution $(\bar{u}, \bar{y})$ to

$$\inf_{(u,y)} \int_{k\tau}^{T+k\tau} \ell(u(t), y(t))\,\mathrm{d}t, \quad \begin{cases} \dot{y}(t) = f(u(t), y(t)) \\ y(k\tau) = y_{MPC}(k\tau). \end{cases}$$

- Define $(u_{MPC}(t), y_{MPC}(t)) = (\bar{u}(t), \bar{y}(t))$, for all $t \in (k\tau, (k+1)\tau)$.

[**Objectives**]
MPC is not just a numerical method for solving long-horizon problems...

- it is also a **real-time** algorithm (assuming the problems of horizon T can be solved instantaneously)
- it is also a feedback mechanism, useful in the context of **disturbances** or model **uncertainties**.

The lecture aims at developing tools (related to dynamic programming) for analysing the optimality of the mechanism.

Methodological comments:

- The continuous-time nature of the system does not play any role, so we will study MPC from the point of view of **discrete-time systems**.
- MPC also allows **constraints** on the control and the state.

2. GENERAL SETTING

[**Data problem**] We introduce now the general setting utilized in the rest of the lecture.

Data of the problem:

- a set Y: the state space
- a set U: the control space
- a subset $\mathbb{Y} \subseteq Y$: the set of feasible states
- a multivalued map $\mathbb{U}$: $y \in \mathbb{Y} \mapsto \mathbb{U}(y) \subseteq U$
- a dynamics f: $Y \times U \to Y$
- a running cost ℓ: $Y \times U \to \mathbb{R}$.

For the moment, we only require that $\ell(u, y) \geq 0$, for any $(u, y) \in U \times Y$.

[**Feasible controls**]

- Given $n \in \mathbb{N} \cup \{\infty\}$, we call **control sequence** (of length n) any $u = (u(k))_{k=0,\dots,n-1} \in U^n$.
- Given an initial condition $x \in \mathbb{Y}$, we call **associated trajectory** $y[u, x] = (y[u, x](k))_{k=0,\dots,n-1} \in Y^{n+1}$ the solution to

$$y[u, x](k+1) = f(y[u, x](k), u(k)), \quad \forall k = 0, \dots n-1$$
$$y[u, x](0) = x.$$

- Given $x \in \mathbb{Y}$, we define the set of **feasible infinite control sequence**

$$\mathbb{U}^\infty(x) = \left\{ u \in U^\infty \;\middle|\; \begin{array}{l} y[u, x](k) \in \mathbb{Y}, \\ u(k) \in \mathbb{U}(y[u, x](k)), \\ \forall k \in \mathbb{N} \end{array} \right\}.$$

[Optimal control problem]

- Given an initial condition x and a control sequence $(u(k))_{k=0,1,\dots} \in U^\infty$, we consider the **cost**
$$J_\infty(u,x) = \sum_{k=0}^{\infty} \ell(u(k), y[u,x](k)) \in \mathbb{R} \cup \{\infty\}.$$
Note that J_∞ can be infinite.
- The **optimal control problem** of interest is:
$$\inf_{u \in \mathbb{U}^\infty(x)} J_\infty(u,x). \qquad (\mathcal{P}_\infty(x))$$

[Stabilization problems]
We restrict ourselves to **stabilization problems**: we assume the existence of a pair (y^*, u^*) such that: $y^* \in \mathbb{Y}$, $u^* \in \mathbb{U}(y^*)$, $y^* = f(u^*, y^*)$ and
$$\ell(u,y) = 0 \Longleftrightarrow (u,y) = (u^*, y^*), \quad \forall (u,y) \in U \times Y.$$
In this framework, the cost function J_∞ steers the state to y^*.
If $y_0 = y^*$, then the solution to $\mathcal{P}$ is the constant sequence $\bar{u}$ equal to u^*, since $\bar{u}$ is feasible and $J_\infty(\bar{u}, y^*) = 0$.

[MPC method] The **MPC method** requires:

- a time horizon $N \in \mathbb{N}$
- a terminal set $\mathbb{Y}_0 \subseteq \mathbb{Y}$
- a terminal cost $F \colon \mathbb{Y}_0 \to \mathbb{R}$.

Given $x \in \mathbb{Y}$, we denote
$$\mathbb{U}^N(x) = \left\{ u \in U^N \;\middle|\; \begin{array}{l} y[u,x](k) \in \mathbb{Y}, \\ u(k) \in \mathbb{U}(y[u,x](k)), \\ \forall k = 0, \dots, N-1 \\ y[u,x](N) \in \mathbb{Y}_0 \end{array} \right\}.$$

[MPC method]
Moreover, given $u \in \mathbb{U}^N(x)$, we define
$$J_N(u,x) = \left(\sum_{k=0}^{N-1} \ell(u(k), y[u,x](k)) \right) + F(y[u,x](N)).$$
The finite-horizon problem utilized in the MPC method is:
$$\inf_{u \in \mathbb{U}^N(x)} J_N(u,x). \qquad (\mathcal{P}_N(x))$$

[MPC method] The MPC method essentially consists in computing a **feedback function** $\mu_N \colon \mathbb{Y} \to U$ in the following fashion: for any $y \in \mathbb{Y}$,

- Find a solution $\bar{u}$ to $\mathcal{P}_N(x)$.
- Set $\mu_N(x) = \bar{u}(0)$.

Then the pair $(u_{\mathrm{MPC}}, y_{\mathrm{MPC}})$ is recursively defined by:
$$\begin{cases} y_{\mathrm{MPC}}(0) = y_0, \\ u_{\mathrm{MPC}}(k) = \mu_N(y_{MPC}(k)), & \forall k = 0,1,\dots \\ y_{\mathrm{MPC}}(k+1) = f(u_{\mathrm{MPC}}(k), y_{\mathrm{MPC}}(k)), & \forall k = 0,1,\dots \end{cases}$$
Equivalently, $y_{\mathrm{MPC}}(k+1) = f_N(y_{\mathrm{MPC}}(k))$, where f_N is defined by
$$f_N(x) = f(\mu_N(x), x).$$

[Viability] For any $N \in \mathbb{N} \cup \{\infty\}$, we consider the set of **feasible initial conditions** (for $\mathcal{P}_N(x)$): $\mathbb{Y}_N = \left\{ x \in \mathbb{Y} \,\middle|\, \mathbb{U}^N(x) \neq \emptyset \right\}$.

Lemma 1. Assume that for any $x \in \mathbb{Y}_0$, there exists $u \in \mathbb{U}(x)$ such that $f(u,x) \in \mathbb{Y}_0$. Then
$$\mathbb{Y}_N \subseteq \mathbb{Y}_{N+1} \subseteq \dots \subseteq \mathbb{Y}_\infty.$$
Moreover, for any $x \in \mathbb{Y}_N$, it holds that
$$f_N(x) = f(\mu_N(x), x) \in \mathbb{Y}_N.$$
Remark: under the assumption of the lemma, if $\mathcal{P}_N(x)$ is feasible in the first step of the method, then it is for all other steps.

[Objectives] The issues related to the existence of a solution to all optimization problems is eluded here. It can be addressed with standard arguments.
We will investigate:

- the **qualitative** behavior of MPC
$\rightarrow$ convergence of y_{MPC} to y^*?
- the **quantitative** behavior of MPC
$\rightarrow$ bound of $J_\infty(u_{\mathrm{MPC}}, x)$?

3. STABILITY ANALYSIS OF DYNAMICAL SYSTEMS

[Stability analysis]

- We forget about the MPC method introduced before and focus on **discrete-time dynamical systems** of the form
$$y(0) = x, \quad y(k+1) = g(y(k)), \quad \forall k = 0,1,\dots$$
for a given initial condition $x \in Y$ and $g \colon X \to X$. We denote the solution by $(y[x](k))_{k=0,1,\dots}$.
- We fix an **equilibrium point** y^* of the dynamics g, that is to say, we assume that $y^* = g(y^*)$.
- We investigate here the **convergence** of $y[x](k)$ to y^*. To this purpose we assume that $\mathbb{Y}$ is metric space and we denote the **distance** of an arbitrary point $y \in Y$ to y^* by $|y|$.

[Comparison functions]
Definition 2. We define here several classes of functions:

- $\mathcal{K} = \left\{ \alpha \colon [0,\infty) \to [0,\infty) \;\middle|\; \begin{array}{l} \alpha \text{ is continuous} \\ \alpha \text{ is strictly increasing} \\ \alpha(0) = 0 \end{array} \right\}$
- $\mathcal{K}_\infty = \left\{ \alpha \in \mathcal{K} \,\middle|\, \alpha(r) \underset{r \to \infty}{\longrightarrow} \infty \right\}$
- $\mathcal{L} = \left\{ \delta \colon \mathbb{N} \to [0,\infty) \;\middle|\; \begin{array}{l} \delta \text{ is strictly decreasing} \\ \delta(t) \underset{t \to \infty}{\longrightarrow} 0 \end{array} \right\}$
- $\mathcal{KL} = \left\{ \beta \colon [0,\infty) \times \mathbb{N} \to \mathbb{R} \;\middle|\; \begin{array}{l} \beta(\cdot, t) \in \mathcal{K}, \ \forall t \in \mathbb{N} \\ \beta(r, \cdot) \in \mathcal{L}, \ \forall r \in [0,\infty) \end{array} \right\}$.

Remark: the t variable in the definition of $\mathcal{L}$ and $\mathcal{KL}$ is usually supposed to lie in $\mathbb{R}$ in the literature.

[**Asymptotic stability**]

Definition 3. We say that y^* is **asymptotically stable** if there exists $\beta \in \mathcal{KL}$ such that for any $x \in \mathbb{Y}$,

$$|y[x](k)| \leq \beta(|x|, k), \quad \forall k \in \mathbb{N}.$$

Remark: this implies that $|y[x](k)| \to 0$ as $k \to \infty$ (not an equivalence).

[**Lyapunov functions**]

Definition 4. We call $V : Y \to [0, \infty)$ a **Lyapunov function** (associated with g) if

- there exists α_1 and $\alpha_2 \in \mathcal{K}_\infty$ such that
$$\alpha_1(|y|) \leq V(y) \leq \alpha_2(|y|), \quad \forall y \in Y$$
- there exists $\alpha_3 \in \mathcal{K}$ such that
$$V(g(y)) \leq V(y) - \alpha_3(|y|), \quad \forall y \in Y.$$

Remark: if V is a Lyapunov function, then $V(y) \geq 0$, for any $y \in Y$. Moreover, $V(y) = 0 \iff y = y^*$ and y^* is the unique equilibrium.

[**Stability under Lyapunov**]

Theorem 5. Assume the existence of a Lyapunov function. Then y^* is asymptotically stable.

Our **objective** for the rest of the lecture: constructing a Lyapunov function for the dynamic system

$$y(k+1) = f_N(y(k)) = f(\mu_N(y(k)), y(k))$$

corresponding to the MPC method.

4. DYNAMIC PROGRAMMING

[**Value function**] For $N \in \mathbb{N} \cup \{\infty\}$, the **value function** $V_N : X \to \mathbb{R} \cup \{\infty\}$ is defined by:

$$V_N(x) = \inf_{u \in \mathbb{U}^N(x)} J_N(u, x).$$

We set $V_N(x) = +\infty$ if $x \notin \mathbb{Y}_N$ (i.e. $\mathbb{U}^N(x)$ is empty).

Theorem 6. For $N \geq 1$ (possibly $N = \infty$), we have

$$V_N(x) = \inf_{u \in \mathbb{U}(x)} \ell(u, x) + V_{N-1}(f(u, x)), \quad \forall x \in \mathbb{Y}. \tag{DP}$$

For $N = 0$, it holds that

$$V_0(x) = \begin{cases} F(x) & \text{if } x \in \mathbb{Y}_0 \\ \infty & \text{otherwise.} \end{cases}$$

[**Dynamic programming**]

Corollary 7. Let $N \geq 1$. Let $x \in \mathbb{Y}_N$.

- Let $\bar{u}$ be a solution to $\mathcal{P}_N(x)$. Then $\bar{u}(0)$ is a solution to (DP). Define $u' \in U^\infty$ by $u'(k) = \bar{u}(k+1)$. Then u' is a solution to $\mathcal{P}_{N-1}(f(u, x))$.
  In particular, $\mu_N(x)$ is a solution to (DP).
- Let u be a solution to (DP). Let u' be a solution to $\mathcal{P}_{N-1}(f(u, x))$. Define $\bar{u} \in U^\infty$ by
$$\bar{u}(0) = u, \quad \bar{u}(k+1) = u'(k), \quad \forall k \in \mathbb{N}.$$
Then $\bar{u}$ is a solution to $\mathcal{P}_N(x)$.

[**From DP to Lyapunov**] A key consequence of the dynamic programming principle is the following:

$$V_{N-1}(f_N(x)) = V_N(x) - \ell(\mu_N(x), x), \quad \forall x \in \mathbb{Y}_N.$$

Observation: this relation is close to the decay condition satisfied by Lyapunov functions.

Can we find **structural assumptions** allowing to use V_N as a Lyapunov function?

[**An abstract result**]

Theorem 8. (1) Assume that there exists $\alpha \in (0, 1]$ such that

$$V_N(f_N(x)) \leq V_N(x) - \alpha\ell(\mu_N(x), x), \quad \forall x \in \mathbb{Y}_N. \tag{ADP}$$

Then the control generated by the MPC method satisfies

$$J_\infty(u^N_{MPC}(x), x) \leq V_N(x)/\alpha.$$

(2) Assume moreover that there exist α_2 and α_3 in $\mathcal{K}_\infty$ such that $V_N(x) \leq \alpha_2(|x|)$ and

$$\inf_{u \in \mathbb{U}(x)} \ell(u, x) \geq \alpha_3(|x|), \quad \forall x \in \mathbb{Y}_N.$$

Then y^* is **asymptotically stable**, for the dynamical system $y(k+1) = f_N(y(k))$.

[**Methodology**] The previous result relies on **non-explicit assumptions**, which remain to be established (under more explicit assumptions...).

We focus on the verification of the **ADP inequality** (for "approximate dynamic programming").

We distinguish two cases:

- MPC **without** terminal cost and terminal constraints: $\mathbb{Y}_0 = \mathbb{Y}$ and $F = 0$.
- MPC **with** terminal cost and constraints.

Each of these two cases requires specific assumptions.

5. MPC WITH TERMINAL COST AND CONSTRAINT

Assumption 1: There exists a map $\kappa : \mathbb{Y}_0 \to U$ satisfying, for any $x \in \mathbb{Y}_0$: $\kappa(x) \in \mathbb{U}(x)$, $f(\kappa(x), x) \in \mathbb{Y}_0$ and $F(f(\kappa(x), x)) \leq F(x) - \ell(\kappa(x), x)$.

Remarks.

- This assumption is in particular satisfied for $\mathbb{Y}_0 = \{y^*\}$, with $\kappa(y^*) = u^*$ and $F = 0$. But the resolution of $\mathcal{P}_N(x)$ is difficult.
- In general, good candidates for F are approximations of V_∞ obtained through linearization techniques.

[**Result**]

Lemma 9. Under Assumption 1, we have

$$V_0(x) \geq \ldots \geq V_{N-1}(x) \geq V_N(x) \geq \ldots V_\infty(x).$$

In particular, $\mathbb{Y}_0 \subseteq \ldots \subseteq \mathbb{Y}_{N-1} \subseteq \mathbb{Y}_N \subseteq \ldots \mathbb{Y}_\infty$.

Corollary 10. Under Assumption 1, V_N satisfies (ADP) with $\alpha = 1$. Therefore

$$J_\infty(u^N_{MPC}(x), x) \leq V_N(x), \quad \forall x \in \mathbb{Y}_N.$$

[**Discussion**]

- The analysis is relatively easy and natural.
- Computation of suitable $\mathbb{Y}_0$ and F is complex.
- Resolution of $\mathcal{P}_N(x)$ possibly difficult.
- Given a feasible initial condition $x \in \mathbb{Y}_\infty$, one possibly needs to have N quite large so that $x \in \mathbb{Y}_N$.
- No general bound on $V_N(x)$ (in comparison with $V_\infty(x)$, yet convergence (w.r.t. N can be achieved).

6. MPC WITHOUT TERMINAL COST AND CONSTRAINT

[**Result**] For any $x \in \mathbb{Y}$, we denote $\ell^*(x) = \inf_{u \in \mathbb{U}(x)} \ell(u, x)$.

Assumption 2

- There is no terminal condition: $\mathbb{Y}_0 = \mathbb{Y}$ and $F = 0$.
- There exist N_0 and $\gamma > 0$ such that

$$V_{N_0}(x) \leq \gamma \ell^*(x), \quad \forall x \in \mathbb{Y}.$$

Lemma 11. Under Assumption 2, it holds that $V_0 \leq V_1 \leq \dots V_\infty$.

Moreover, the ADP inequality holds for $N = N_0$ and

$$\alpha_{N_0} = 1 - \frac{\gamma^2}{N_0 - 1}.$$

[**Discussion**]

- Resolution of $\mathcal{P}_N(x)$ easier without terminal condition (used in practice).
- The method also requires N_0 to be sufficiently large (so that $\alpha_{N_0} > 0$).
- Refinement are possible (i.e. sharper estimates of α_{N_0} on coefficients γ which depend on N_0).
- Proof of existence of γ doable on a case-by-case basis.